

Hyunsoo Kim

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EDUCATION

The University of Texas at Austin

Ph.D. in Electrical and Computer Engineering

Austin, TX

Sep 2026 – Present

- Co-advised by Prof. Guandao Yang and Prof. Diana Marculescu
- Research: Leveraging image/video generation models for visual understanding and perception

Korea University

M.S. in Artificial Intelligence

Seoul, Republic of Korea

Sep 2024 – Aug 2026

- Co-advised by Prof. Donghyun Kim and Prof. Suhyun Kim

PUBLICATIONS

Correlation-Weighted Multi-Reward Optimization for Compositional Generation

ECCV 2026

- Jungmyung Wi, **Hyunsoo Kim**, Donghyun Kim
- Proposed correlation-based reward reweighting for multi-concept T2I generation with GRPO fine-tuning

Self-Improving Diffusion Classifiers with Minority Preference Optimization

ECCV 2026

- **Hyunsoo Kim***, Jungmyung Wi*, Soobin Um, Donghyun Kim, Suhyun Kim (*: equal contribution)
- Introduced MiPO: self-improving framework expanding diffusion coverage in low-density regions via GRPO with minority preference rewards, improving zero-shot classification by up to 3.8%

Diffusion-assisted Data-free Image Synthesis with Domain-Class Alignment

ICML 2025

- Yujin Kim*, **Hyunsoo Kim***, Hyunwoo J. Kim, Suhyun Kim (*: equal contribution)
- Proposed DDIS: leveraging T2I diffusion models for data-free knowledge distillation with domain-class alignment

Difference Inversion for Image Analogy Generation

CVPR 2025

- **Hyunsoo Kim**, Donghyun Kim, Suhyun Kim
- Token-consistency-based DDIM inversion to interpolate and isolate visual differences for image analogy

PREPRINTS

Safety-Aware Image-to-Image Translation without Paired Data – **Hyunsoo Kim**, Wonjun Lee, Donghyun Kim, Suhyun Kim

Consistency-Preserving Concept Erasure via Unsafe-Safe Pairing – Yongwoo Kim*, Sungmin Cha*, **Hyunsoo Kim**, et al.

EXPERIENCE

Korea Institute of Science and Technology (KIST)

Graduate Researcher (Advisor: Suhyun Kim)

Seoul, Republic of Korea

Sep 2024 – Aug 2026

- Conducted research on image analogy generation, safe image translation, and self-improving diffusion classifiers
- Published 2 first-author papers at ECCV 2026 and CVPR 2025; 1 co-first-author paper at ICML 2025

Korea Institute of Science and Technology (KIST)

Undergraduate Research Intern (Advisor: Suhyun Kim)

Seoul, Republic of Korea

Mar 2024 – Jun 2024

- Researched data-free image synthesis by optimizing text embeddings on text-to-image diffusion models

Medical Image Innovation Lab, Seoul National University

Undergraduate Intern (Advisor: Wonjin Lee)

Seoul, Republic of Korea

Dec 2023 – Feb 2024

- Researched conditional generation and inpainting in medical imaging using diffusion models

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Bash

ML Frameworks: PyTorch, Hugging Face (Diffusers, Accelerate, TRL), DeepSpeed, JAX

Tools: Git, Docker, Weights & Biases, SLURM, Linux, $\LaTeX$

Research Areas: Diffusion Models, Flow Matching, RLHF/GRPO, Image/Video Generation, Zero-shot Classification, AI Safety